# Ian Waudby-Smith

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## Academic Positions

**University of California, Berkeley** — **Berkeley, CA**
*Miller Postdoctoral Fellow* — *2024–2027*

## Education

**Carnegie Mellon University** — **Pittsburgh, PA**
*PhD, Statistics* — *2024*
Thesis: *Advances in anytime-valid sequential inference*
Recipient of the Umesh K. Gavaskar Memorial Dissertation Award

**University of Waterloo** — **Waterloo, Ontario**
*BMath, Pure Mathematics & Statistics* — *2018*
Dean's Honours List

## Publications

[1] Chase Mathis and **Ian Waudby-Smith**. Confidence horizons. *Under review at the Journal of the Royal Statistical Society, Series B.*

[2] Keegan Harris, Brian W. Lee, **Ian Waudby-Smith**, Philip Amortila, Nika Haghtalab, and Michael I. Jordan. Post-training at the edge of detectability: A game-theoretic approach to fine-tuning. *Under review at the International Conference on Learning Representations (ICLR)*, 2026.

[3] ($\alpha\beta$) Jiahao Ming, Aaditya Ramdas, Yi Shen, Ruodu Wang, and **Ian Waudby-Smith**. Gaffke's confidence interval for the mean of bounded data is inadmissible but asymptotically efficient. *arXiv preprint arXiv:2607.18661.*

[4] ($\alpha\beta$) Jiahao Ming, Aaditya Ramdas, Yi Shen, Ruodu Wang, and **Ian Waudby-Smith**. Combining e-values via demi-supermartingales. *Under review at the Annals of Applied Probability.*

[5] Ricardo J. Sandoval*, **Ian Waudby-Smith***, and Michael I. Jordan. Multi-armed sequential hypothesis testing by betting. *arXiv preprint arXiv:2603.17925.*

[6] Ben Chugg*, Etienne Gauthier*, Michael I. Jordan, Aaditya Ramdas, and **Ian Waudby-Smith**. Post-hoc large sample statistical inference. *Under review at the Journal of the Royal Statistical Society, Series B*, 2026.

[7] Ricardo J. Sandoval, Sivaraman Balakrishnan, Avi Feller, Michael I. Jordan, and **Ian Waudby-Smith**. On nonasymptotic confidence intervals for treatment effects in randomized experiments. *Major revision at Biometrika.*

[8] **Ian Waudby-Smith**, Ricardo J. Sandoval, and Michael I. Jordan. Instance-log-optimality of portfolio-based e-processes and their sequential hypothesis tests. *R&R at the Annals of Statistics*, 2025.

[9] **Ian Waudby-Smith**, Martin Larsson, and Aaditya Ramdas. Nonasymptotic and distribution-uniform Komlós-Major-Tusnády approximation. *Major revision at Probability Theory and Related Fields*.

[10] **Ian Waudby-Smith**, Edward H. Kennedy, and Aaditya Ramdas. Distribution-uniform large sample anytime-valid inference and the Robbins–Siegmund distributions. *Major revision at Biometrika*.

[11] ($\alpha\beta$) Johannes Ruf and **Ian Waudby-Smith**. Concentration inequalities for strong laws and laws of the iterated logarithm. *Bulletin of the London Mathematical Society (to appear)*, 2026.

[12] **Ian Waudby-Smith**, Martin Larsson, and Aaditya Ramdas. Distribution-uniform strong laws of large numbers. *The Annals of Applied Probability (to appear)*, 2026.

[13] **Ian Waudby-Smith**, David Arbour, Ritwik Sinha, Edward H. Kennedy, and Aaditya Ramdas. Time-uniform central limit theory and asymptotic confidence sequences. *The Annals of Statistics*, 52(6):2613–2640, 2024.

[14] **Ian Waudby-Smith**, Lili Wu, Aaditya Ramdas, Nikos Karampatziakis, and Paul Mineiro. Anytime-valid off-policy inference for contextual bandits. *ACM/IMS Journal of Data Science* ***(Invited contribution)***, 1(3):1–42, 2024.

[15] **Ian Waudby-Smith** and Aaditya Ramdas. Estimating means of bounded random variables by betting. *Journal of the Royal Statistical Society Series B: Statistical Methodology* ***(Discussion paper)***, 86(1):1–27, 2024.

[16] **Ian Waudby-Smith**, Zhiwei Steven Wu, and Aaditya Ramdas. Extensions of randomized response for private confidence sets. *International Conference on Machine Learning* ***(Oral presentation)***, 2023.

[17] Akash V. Maharaj, Ritwik Sinha, David Arbour, **Ian Waudby-Smith**, Simon Z. Liu, Moumita Sinha, Raghavendra Addanki, Aaditya Ramdas, Manas Garg, and Viswanathan Swaminathan. Anytime-valid confidence sequences in an enterprise A/B testing platform. *The ACM World Wide Web Conference*, 2024.

[18] **Ian Waudby-Smith**, Philip B. Stark, and Aaditya Ramdas. RiLACS: Risk limiting audits via confidence sequences. In *International Joint Conference on Electronic Voting* ***(Best paper award)***, pages 124–139. Springer, 2021.

[19] **Ian Waudby-Smith** and Aaditya Ramdas. Confidence sequences for sampling without replacement. *Advances in Neural Information Processing Systems* ***(Spotlight paper)***, 33:20204–20214, 2020.

[20] **Ian Waudby-Smith**, A Simon Pickard, Feng Xie, and Eleanor M. Pullenayegum. Using both time tradeoff and discrete choice experiments in valuing the EQ-5D: Impact of model misspecification on value sets. *Medical Decision Making*, 2020.

[21] **Ian Waudby-Smith**, Nam Tran, Joel A. Dubin, and Joon Lee. Sentiment in nursing notes as an indicator of out-of-hospital mortality in intensive care patients. *PloS one*, 13(6), 2018.

($\alpha\beta$) Alphabetical ordering.

*Co-lead author.

## Teaching Experience

**Carnegie Mellon University** **Pittsburgh, PA**
*Graduate Teaching Assistant* *2019–22*

- 36-708: Statistical Methods in Machine Learning (x2)
- 36-462: Data Mining
- 36-401: Modern Regression
- 36-731: Foundations of Causal Inference
- 36-732: Modern Causal Inference
- 10-880: Game-theoretic Probability, Statistics, and Learning

## Other Experience

**Google Research** **New York, NY**
*Student Researcher* *Jun–Aug 2023*
Mentors: Jean Pouget-Abadie & Jennifer Brennan

**Microsoft Research** **Redmond, WA**
*Research Intern* *May–Aug 2022*
Mentor: Paul Mineiro

**Adobe Research** **San Jose, CA**
*Research Intern* *Jun–Aug 2020*
Mentors: David Arbour & Ritwik Sinha

**The Hospital for Sick Children (SickKids)** **Toronto, Ontario**
*Research Student* *Apr–Aug 2019*
Mentor: Eleanor Pullenayegum

**Health Data Science Lab, University of Waterloo** **Waterloo, Ontario**
*Research Assistant* *2016–18*
Mentors: Joel Dubin & Joon Lee

**Department of Statistics, University of Waterloo** **Waterloo, Ontario**
*Research Assistant* *Apr–Aug 2017*
Mentor: Pengfei Li

**Department of Pure Mathematics, University of Waterloo** **Waterloo, Ontario**
*Research Assistant* *Jan - Apr 2017*
Mentor: Yu-Ru Liu

**Cancer Care Ontario** **Toronto, Ontario**
*Student Analyst* *Jan–Apr 2016*
Mentor: Zhihui (Amy) Liu

## Professional Service

**Thesis committee member**

- Tyron Lardy (PhD in Statistics, Leiden University)
- Noah Liniger (MSc in Data Science, ETH Zurich)

**Reviewer**

- Annals of Statistics
- Bernoulli
- Biometrika
- Electronic Journal of Probability
- Electronic Journal of Statistics
- Journal of the American Statistical Association
- Journal of the Royal Statistical Society, Series B
- Management Science
- New England Journal of Data Science
- Sankhya A.
- Symposium on Theory of Computing (STOC)

**University of California, Berkeley** **Berkeley, CA**
*Volunteer*

- Miller symposium organizing committee
- Incoming Miller fellow initiative

**Carnegie Mellon University** **Pittsburgh, PA**
*Volunteer*

- Organizer of the Statistical Machine Learning Reading Group (SMLRG)
- Women in Data Science (WiDS) conference volunteer
- Computing committee student representative
- Incoming PhD student mentor

## Awards

**Institute of Mathematical Statistics** **Berkeley, CA**
*New Researcher Travel Award* *2026*

**Miller Institute for Basic Research in Science** **Berkeley, CA**
*Miller fellowship* *2024–'27*

**Carnegie Mellon University Department of Statistics and Data Science** **Pittsburgh, PA**
*2024*
*Umesh K. Gavaskar Best Dissertation Award*

**Statistical Society of Canada** **St. John's, NL**
*Probability Section Student Research Presentation Award* *2024*

**Amazon Science** **Pittsburgh, PA**
*Graduate Research Fellowship* *2023*

**University of Waterloo** **Waterloo, Ontario**
*Waterloo Statistics Student Conference Presentation Award* *2022*

**Carnegie Mellon University Department of Statistics and Data Science** **Pittsburgh, PA**
*2021*
*Teaching Assistant of the Year*

**Adobe Research** **Pittsburgh, PA**
*PhD Research Gift* *2020*

**University of Waterloo** **Waterloo, Ontario**
*David Johnston International Experience Award* *2018*

**The Natural Sciences and Engineering Research Council of Canada** **Waterloo, Ontario**
*NSERC Undergraduate Student Research Award* *2017*

**University of Waterloo** **Waterloo, Ontario**
*President's Research Award* *2016–17*

**University of Waterloo** **Waterloo, Ontario**
*President's Scholarship of Distinction* *2014*

## Presentations

### *Short courses*

**MBZUAI** **Abu Dhabi, UAE**
*A brief introduction to game-theoretic, safe, anytime-valid inference* *2025*
Mini course consisting of 4 lectures

**ERC OCEAN retreat** **Venice, Italy**
*A brief introduction to game-theoretic, safe, anytime-valid inference* *2024*
Mini course consisting of 3 lectures

### *Department seminars*

**Department of Statistics at the London School of Economics** **London, UK**
*Log-optimality and multi-armed sequential hypothesis testing* *2026*

**Department of Statistics and Actuarial Science at the University of Waterloo** **Waterloo, Ontario**
*Log-optimality and multi-armed sequential hypothesis testing* *2026*

**Department of Statistics at University of California, Davis** **Davis, CA**
*Log-optimality and multi-armed sequential hypothesis testing* *2026*

**Department of Statistics at Iowa State University** **Ames, IA**
*Anytime-valid inference and uniform central limit theory* *2026*

**Division of Biostatistics at the University of California, Berkeley** **Berkeley, CA**
*Anytime-valid off-policy inference for contextual bandits* *2025*

**Department of Statistics at Stanford University** **Stanford, CA**
*Anytime-valid inference and uniform central limit theory* *2025*

**Statistics Seminar at the University of Copenhagen** **Copenhagen, Denmark**
*Anytime-valid off-policy inference for contextual bandits* *2023*

## *Invited talks at conferences / workshops / research groups*

**Institute for Operations Research and the Management Sciences (INFORMS)** — **San Francisco, CA**, *2026*
*Multi-armed log-optimality and sequential testing*

**Whendee L. Silver lab at the University of California, Berkeley** — **Berkeley, CA**, *2026*
*Anytime-valid inference for the environmental sciences*

**Inria/Sierra Seminar** — **Paris, France**, *2026*
*Combining e-values via demi-supermartingales*

**Cornell University Rising Stars in Statistics and Data Science Workshop** — **Ithaca, NY**, *2026*
*Advances in asymptotic anytime-valid inference*

**Joint Statistical Meetings (JSM)** — **Boston, MA**, *2026*
*Asymptotic anytime-valid inference*

**Adaptive Experimentation Workshop at University College London (Gatsby)** — **London, UK**, *2026*
*Log-optimality and multi-armed sequential hypothesis testing*

**International Workshop on Sequential Methodologies** — **Washington, D.C.**, *2026*
*Asymptotic anytime-valid statistical inference*
Recipient of the IMS New Researcher Travel Award

**Stanford Causal Inference Reading Group (host: Vasilis Syrgkanis)** — **Stanford, CA**, *2026*
*On nonasymptotic confidence intervals for treatment effects in randomized experiments*

**National University of Singapore Young Mathematical Scientists Forum** — **Singapore**, *2025*
*Log-optimality of e-processes and sequential tests*

**Stanford Data Driven Seminar (host: Stefan Wager)** — **Stanford, CA**, *2025*
*Log-optimality of e-processes and sequential tests*

**MBZUAI-Berkeley Workshop** — **Abu Dhabi, UAE**, *2025*
*Anytime-valid off-policy inference for contextual bandits*

**Inria/Sierra Seminar** — **Paris, France**, *2025*
*Anytime-valid inference and uniform central limit theory*

**International Seminar on Selective Inference** — **Virtual**, *2024*
*$\mathcal{P}$-uniform anytime-valid inference and conditional independence testing without Model-X*

**CLIMB Workshop** — **Berkeley, CA**, *2024*
*Election audits via anytime-valid inference*

**Workshop on Game-Theoretic Statistical Inference** — **Oberwolfach, Germany**, *2024*
*$\mathcal{P}$-uniform anytime-valid inference and conditional independence testing without Model-X*

**Fienberg Student Research Workshop at Carnegie Mellon University** — **Pittsburgh, PA**, *2024*
*Election audits via anytime-valid inference*

**Joint Statistical Meetings** **Toronto, Ontario**
*Anytime-valid off-policy inference for contextual bandits* *2023*

**International Conference on Machine Learning** **Honolulu, HI**
*Extensions of randomized response for private confidence sets* *2023*

**Centrum Wiskunde & Informatica (hosts: Peter Grünwald & Wouter Koolen)** **Amsterdam, Netherlands**
*Anytime-valid off-policy inference for contextual bandits* *2023*

**University of Toronto / The Vector Institute (host: Dan Roy)** **Toronto, Ontario**
*Anytime-valid off-policy inference for contextual bandits* *2023*

**Copenhagen Causality Lab, University of Copenhagen** **Virtual**
*Asymptotic confidence sequences for anytime-valid causal inference* *2023*

**Conference on Digital Experimentation (CODE@MIT)** **Cambridge, MA**
*Asymptotic confidence sequences for anytime-valid causal inference* *2022*

**Microsoft Research Reinforcement Learning Discussion Group** **Virtual**
*Anytime-valid contextual bandit inference* *2022*

**Computing and Mathematical Sciences, California Institute of Technology (host: Houman Owhadi)** **Virtual**
*A brief introduction to safe, anytime-valid inference (SAVI)* *2022*

**Microsoft Research** **Virtual**
*A brief introduction to safe, anytime-valid inference (SAVI)* *2022*

**Safe, Anytime-Valid Inference (SAVI) Workshop** **Eindhoven, Netherlands**
*Time-uniform central limit theory and anytime-valid causal inference* *2022*

**Carnegie Mellon University Computer Science Theory Lunch** **Pittsburgh, PA**
*Estimating means of bounded random variables by betting* *2021*

**International Seminar on Distribution-Free Statistics** **Virtual**
*Estimating means of bounded random variables by betting* *2021*

**E-Vote-ID: The International Conference for Electronic Voting** **Virtual**
*RiLACS: Risk-limiting audits via confidence sequences* *2021*

**Spotify Experimentation Platform Team** **Virtual**
*Doubly robust confidence sequences for sequential causal inference* *2021*

**Vinted Science and Analytics Meetup** **Virtual**
*Doubly robust confidence sequences for sequential causal inference* *2021*

### *Contributed conference talks*

**International Conference on Statistics and Data Science (ICSDS)** **Seville, Spain**
*Concentration inequalities for strong laws of large numbers* *2025*

**Statistical Society of Canada meeting** **St. John's, NL**
*Distribution-uniform strong laws of large numbers* *2024*
Recipient of the Probability Section's Student Presentation Award

**International Conference on Statistics and Data Science (ICSDS)** — **Lisbon, Portugal**, *2023*
*Distribution-uniform anytime-valid inference*

**Waterloo Student Conference in Statistics, Actuarial Science, and Finance** — **Waterloo, Ontario**, *2022*
*Estimating means of bounded random variables by betting*
Recipient of the Student Presentation Award

**TPDP: Theory and Practice of Differential Privacy Workshop** — **Baltimore, MD**, *2022*
*Locally private nonparametric confidence intervals and sequences*

**Statistical Society of Canada (SSC) Annual Meeting** — **Virtual**, *2022*
*Time-uniform central limit theory and anytime-valid causal inference*

**NeurIPS Workshop on Causal Inference Challenges in Sequential Decision Making** — **Virtual**, *2021*
*Time-uniform central limit theory and anytime-valid causal inference*

**Joint Statistical Meetings (JSM)** — **Virtual**, *2021*
*Doubly robust confidence sequences for sequential causal inference*

**Joint Statistical Meetings (JSM)** — **Virtual**, *2020*
*Confidence sequences for sampling without replacement*

**Statistical Society of Canada (SSC) Annual Meeting** — **St. Catherines, Ontario**, *2016*
*Multi-state models for chronic kidney disease prevalence projections in Ontario*

Last updated: October 2, 2026